Network Motifs: Simple Building Blocks of Complex Networks

R. Milo *et al. Science* **298**, 824 (2002).

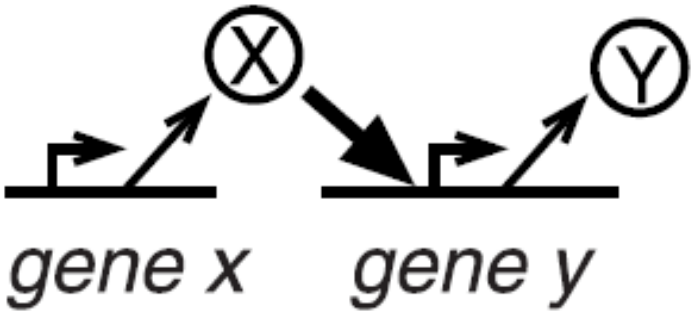
Beyond Global Features: Network Motifs

- **Network Motifs**: Patterns of interconnections occurring in complex networks at numbers that are significantly higher than those in randomized networks

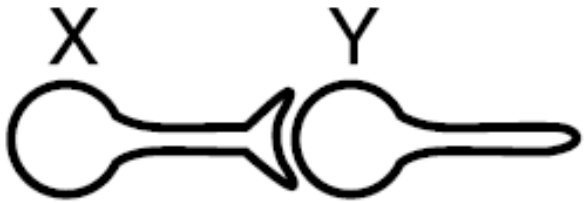
- Do networks from different domains (such as biochemistry, neurobiology, ecology, and engineering)?

X→Y represents

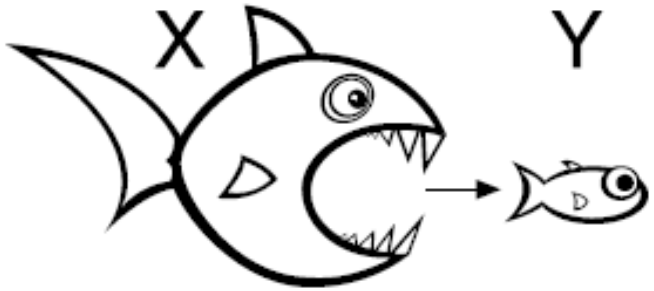
transcription network



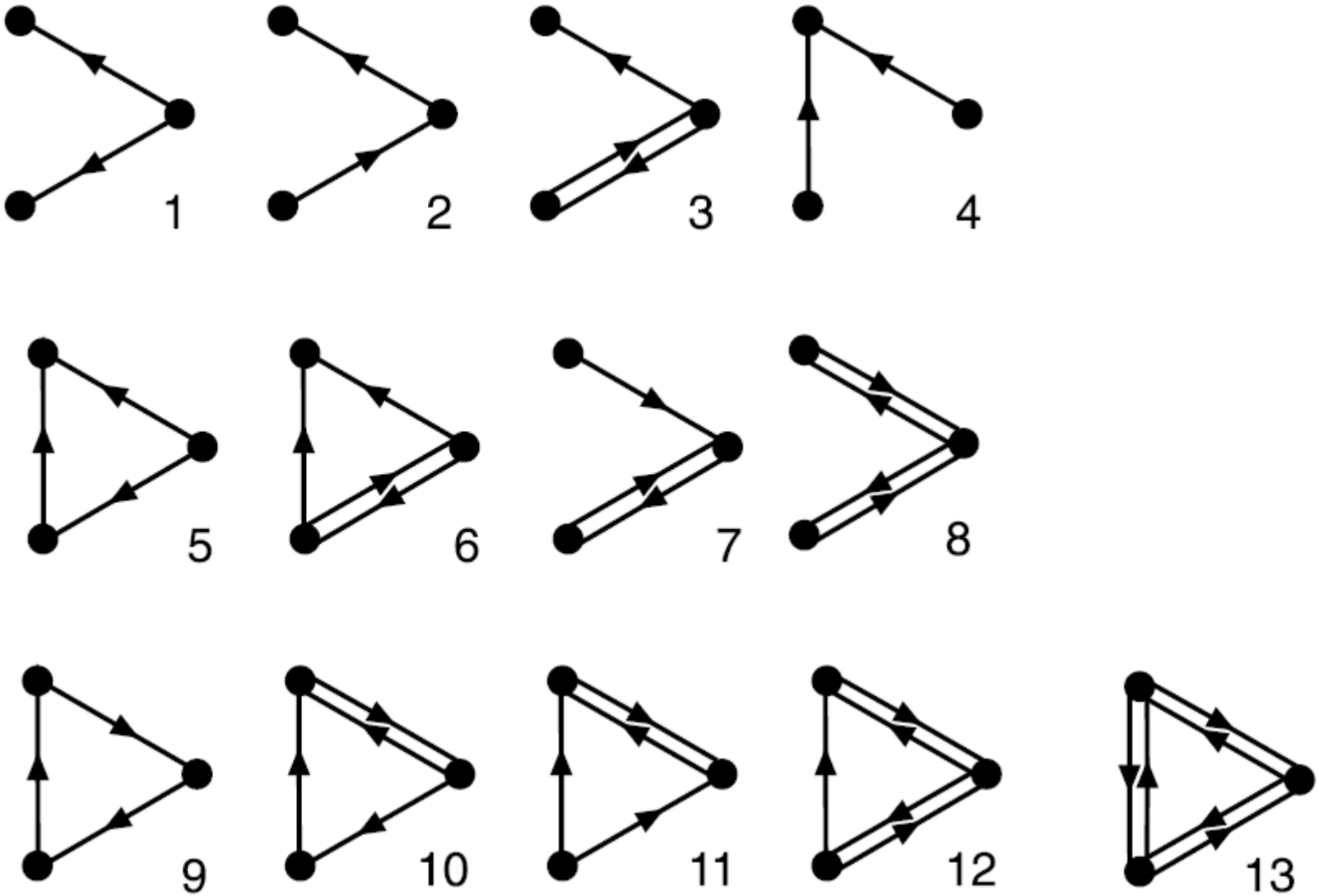
neuron synaptic connection network



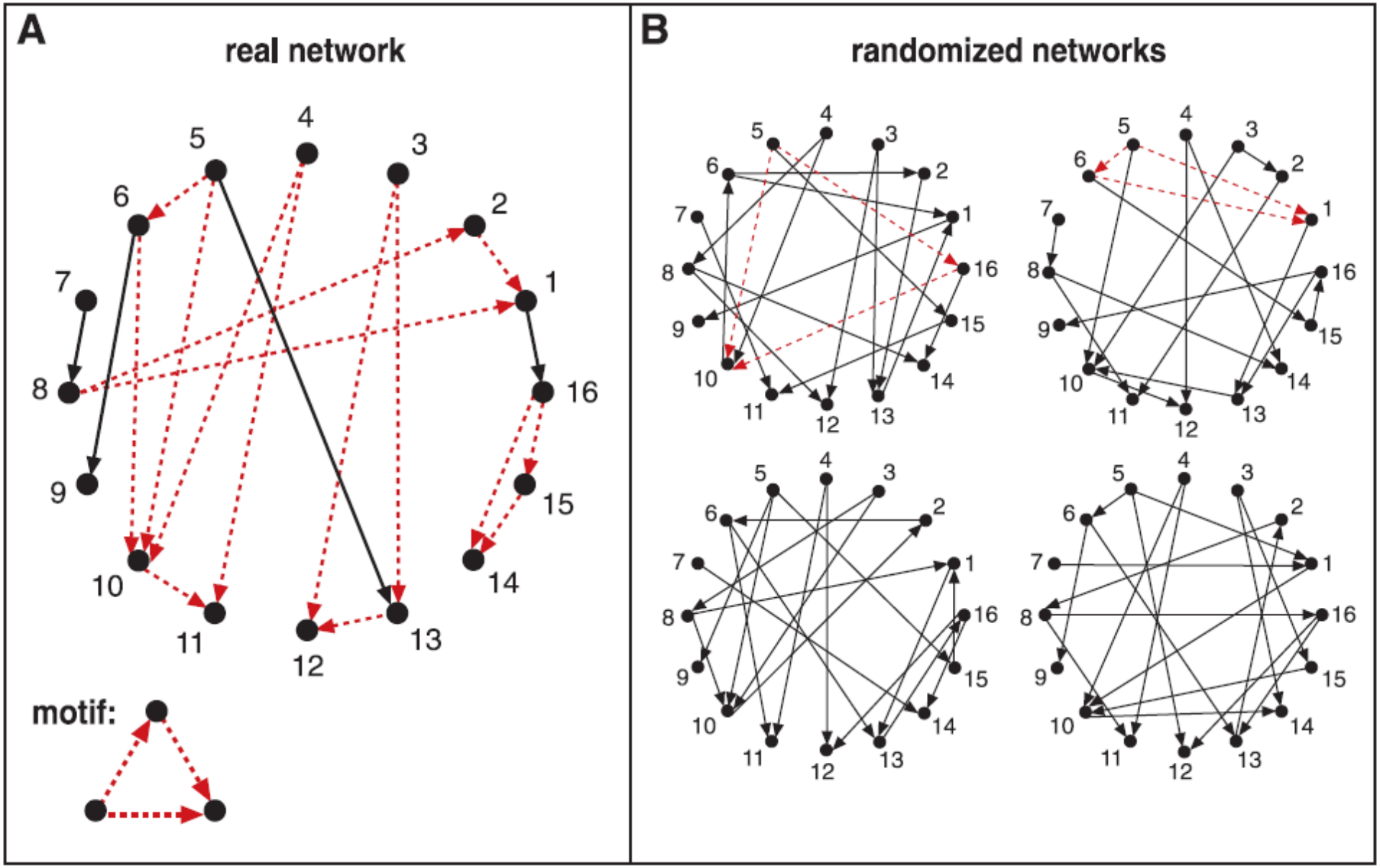
ecological food web



13 types of three-node connected subgraphs.



Network Motif Detection



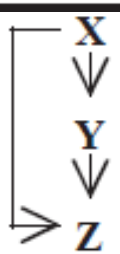
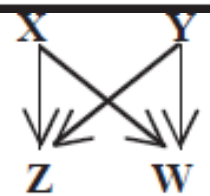
Network Motif Detection

- For each motif, the numbers of appearances in the real network (N_{real}) and in the randomized networks ($N_{\text{rand}} \pm \text{SD}$) were computed.

- Qualitative measure of statistical significance: $Z_{score} = \frac{N_{real} - N_{rand}}{SD}$

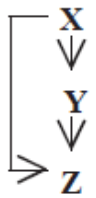
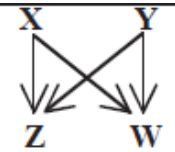
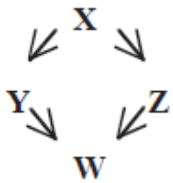
Network Motifs: Transcriptional Networks

- Transcription networks are biochemical networks responsible for regulating the expression of genes in cells.
- Directed graphs. Nodes → genes.
- Edges are directed from a gene that encodes for a transcription factor protein to a gene transcriptionally regulated by that transcription factor.
- An eukaryote (the yeast, *Saccharomyces cerevisiae*) and a bacterium (*Escherichia coli*).

Network	Nodes	Edges	N_{real}	$N_{rand} \pm SD$	Z score	N_{real}	$N_{rand} \pm SD$	Z score
Gene regulation (transcription)					**Feed-forward loop**			**Bi-fan**
E. coli	424	519	40	7 ± 3	10	203	47 ± 12	13
*S. cerevisiae**	685	1,052	70	11 ± 4	14	1812	300 ± 40	41

Network Motifs: Neuronal Connectivity Networks

- Neuronal connectivity network of the nematode *Caenorhabditis elegans*.
- Nodes → neurons (or neuron classes)
- Edges → synaptic connections between the neurons.

Network	Nodes	Edges	N_{real}	$N_{rand} \pm SD$	Z score	N_{real}	$N_{rand} \pm SD$	Z score	N_{real}	$N_{rand} \pm SD$	Z score
Neurons					**Feed-forward loop**			**Bi-fan**			**Bi-parallel**
C. elegans†	252	509	125	90 ± 10	3.7	127	55 ± 13	5.3	227	35 ± 10	20

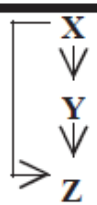
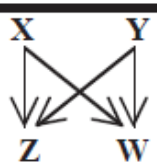
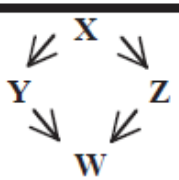
Network Motifs: Food Webs

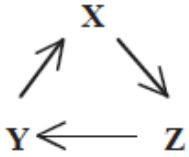
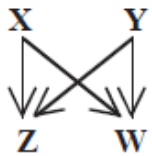
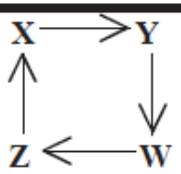
- Trophic Interactions
- Nodes → species
- Edges are directed from a node representing a predator to the node representing its prey.
- Seven distinct ecosystems (aquatic and terrestrial habitats).
- Each of the food webs displayed one or two three-node network motifs and one to five four-node network motifs.

Network	Nodes	Edges	N_{real}	$N_{rand} \pm SD$	Z score	N_{real}	$N_{rand} \pm SD$	Z score
Food webs				**X** ⇓ **Y** ⇓ **Z**	**Three chain**	**X** ↙ ↘ **Y** ↘ ↙ **Z** **W**		**Bi-parallel**
Little Rock	92	984	3219	3120 ± 50	2.1	7295	2220 ± 210	25
Ythan	83	391	1182	1020 ± 20	7.2	1357	230 ± 50	23
St. Martin	42	205	469	450 ± 10	NS	382	130 ± 20	12
Chesapeake	31	67	80	82 ± 4	NS	26	5 ± 2	8
Coachella	29	243	279	235 ± 12	3.6	181	80 ± 20	5
Skipwith	25	189	184	150 ± 7	5.5	397	80 ± 25	13
B. Brook	25	104	181	130 ± 7	7.4	267	30 ± 7	32

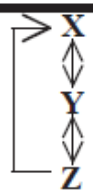
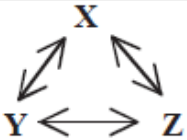
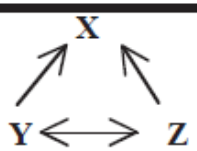
Network Motifs: Electronic Circuits

ISCAS89 benchmark set of sequential logic electronic circuits we re analyzed.
Nodes → logic gates and flip-flops
The motifs separate the circuits into classes that correspond to the circuit's functional description.
The forward logic chips share the feedforward loop, bi-fan, and bi-parallel motifs (similar to those found in the genetic and neuronal networks.)

Network	Nodes	Edges	N_{real}	$N_{rand} \pm SD$	Z score	N_{real}	$N_{rand} \pm SD$	Z score	N_{real}	$N_{rand} \pm SD$	Z score
Electronic circuits (forward logic chips)					**Feed-forward loop**			**Bi-fan**			**Bi-parallel**
s15850	10,383	14,240	424	2 ± 2	285	1040	1 ± 1	1200	480	2 ± 1	335
s38584	20,717	34,204	413	10 ± 3	120	1739	6 ± 2	800	711	9 ± 2	320
s38417	23,843	33,661	612	3 ± 2	400	2404	1 ± 1	2550	531	2 ± 2	340
s9234	5,844	8,197	211	2 ± 1	140	754	1 ± 1	1050	209	1 ± 1	200
s13207	8,651	11,831	403	2 ± 1	225	4445	1 ± 1	4950	264	2 ± 1	200

Network	Nodes	Edges	N_{real}	$N_{rand} \pm SD$	Z score	N_{real}	$N_{rand} \pm SD$	Z score	N_{real}	$N_{rand} \pm SD$	Z score
Electronic circuits (digital fractional multipliers)					**Three-node feedback loop**			**Bi-fan**			**Four-node feedback loop**
s208	122	189	10	1 ± 1	9	4	1 ± 1	3.8	5	1 ± 1	5
s420	252	399	20	1 ± 1	18	10	1 ± 1	10	11	1 ± 1	11
s838‡	512	819	40	1 ± 1	38	22	1 ± 1	20	23	1 ± 1	25

Network Motifs: World-wide Web

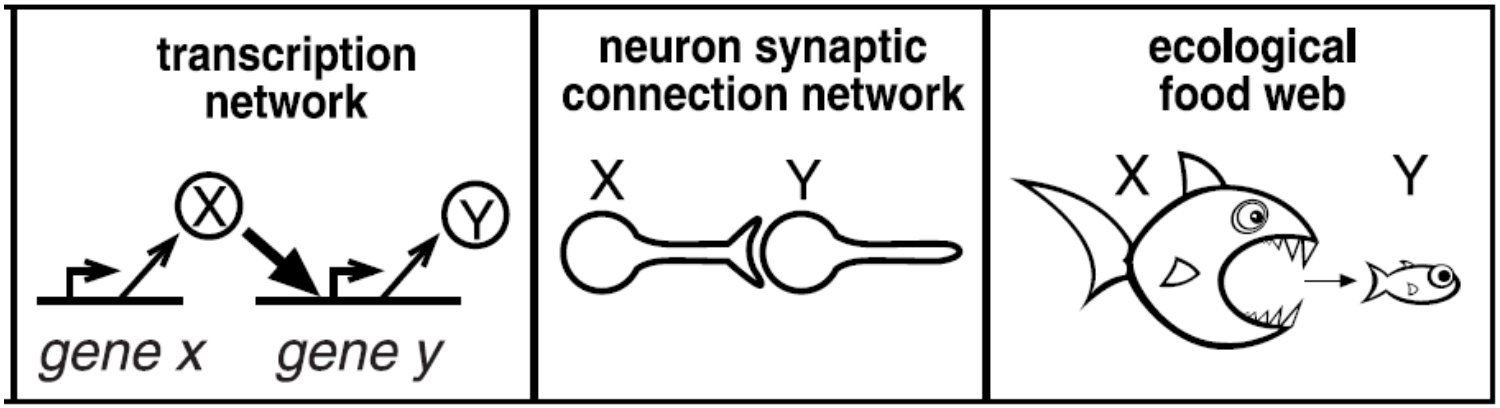
Network	Nodes	Edges	N_{real}	$N_{rand} \pm SD$	Z score	N_{real}	$N_{rand} \pm SD$	Z score	N_{real}	$N_{rand} \pm SD$	Z score
World Wide Web					**Feedback with two mutual dyads**			**Fully connected triad**			**Uplinked mutual dyad**
nd.edu§	325,729	1.46e6	1.1e5	2e3 ± 1e2	800	6.8e6	5e4±4e2	15,000	1.2e6	1e4 ± 2e2	5000

Why common motifs?

- Two of these motifs (feedforward loop and bi-fan) were found in both the transcriptional gene regulation networks as well as neuronal networks.
- This similarity in motifs may point to a **fundamental similarity in the design constraints** of the two types of networks.
- Both networks function to carry information from sensory components (sensory neurons/transcription factors regulated by biochemical signals) to effectors (motor neurons/structural genes).
- The **feedforward loop motif** common to both types of networks may play a functional role in information processing.
- One possible **function** of this circuit is **to activate output only if the input signal is persistent and to allow a rapid deactivation when the input goes off**.
- Indeed, many of the input nodes in the neural feedforward loops are sensory neurons, which may require this type of information processing to reject transient input fluctuations that are inherent in a variable or noisy environment.

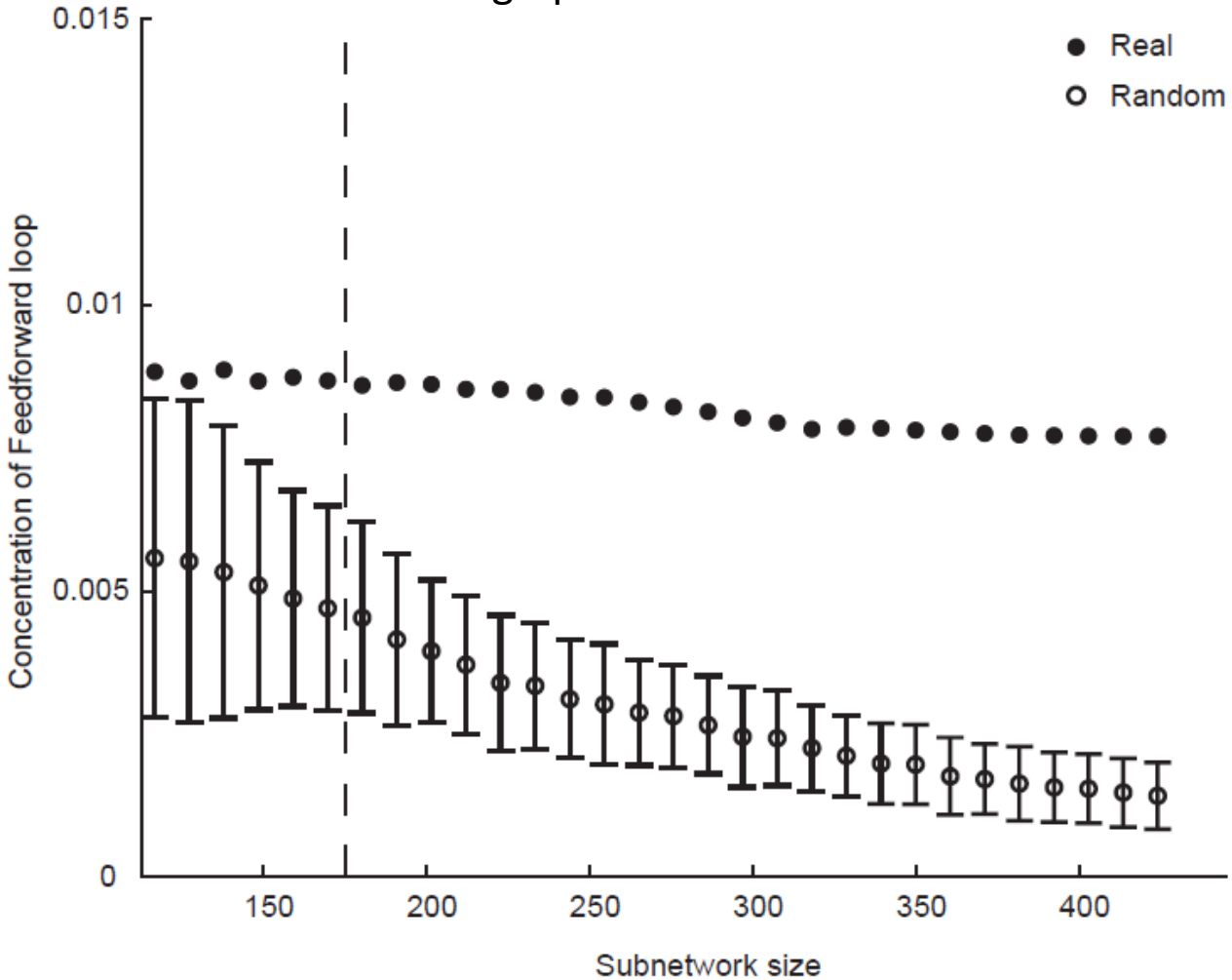
Why specific network motifs?

- These results suggests that **motifs can define broad classes of networks**, each with specific types of elementary structures.
- The motifs reflect the **underlying processes that generated each type of network**.
- For example, food webs evolve to allow a **flow of energy** from the bottom to the top of food chains, whereas gene regulation and neuron networks evolve to **process information**.
- Information processing seems to give rise to significantly different structures than does energy flow.



Concentration of Feedforward Motif

Concentration (*C*) of the feedforward loop motif in real and randomized subnetworks of the *E. coli* transcription network.
C is the number of appearances of the motif divided by the total number of appearances of all connected three-node subgraphs.



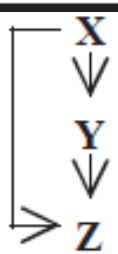
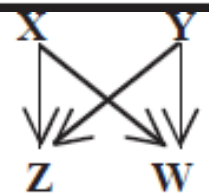
Motifs: Overrepresented patterns of connectivity

- Motifs → Functional specification?
- Good software design vs. Bad software design
- Brain networks: Normally functioning vs. Disease Conditions?
- Ecosystems: Stable vs. Fragile

Z-score

- Compute the Z-score.

$$Z_{score} = \frac{N_{real} - N_{rand}}{SD}$$

Network	Nodes	Edges	N_{real}	$N_{rand} \pm SD$	Z score	N_{real}	$N_{rand} \pm SD$	Z score
Gene regulation (transcription)					**Feed-forward loop**			**Bi-fan**
E. coli	424	519	40	7 ± 3	10	203	47 ± 12	13
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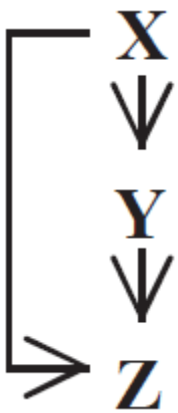
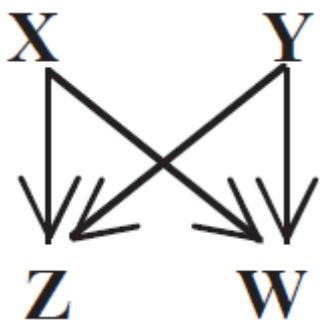
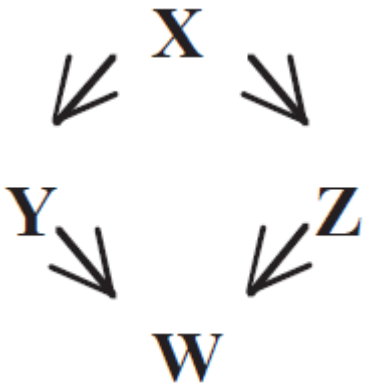
Inter-convertibility of Motifs



Feedforward Loop

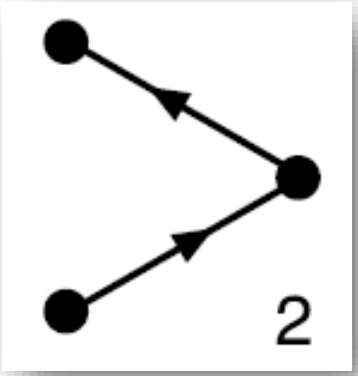


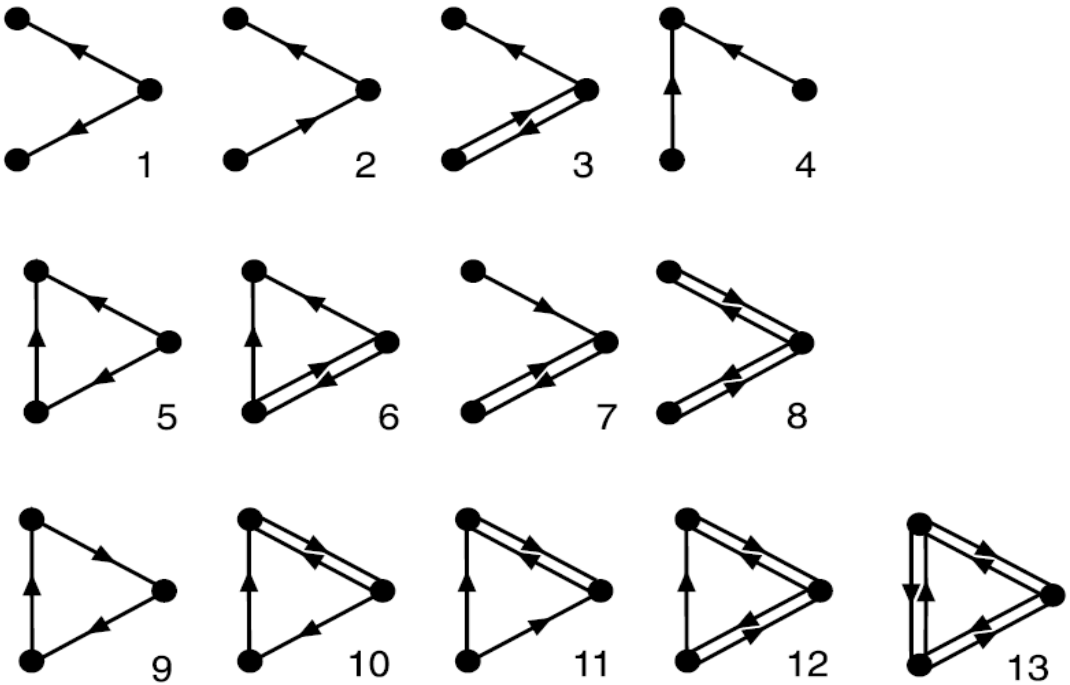
Bi-fan

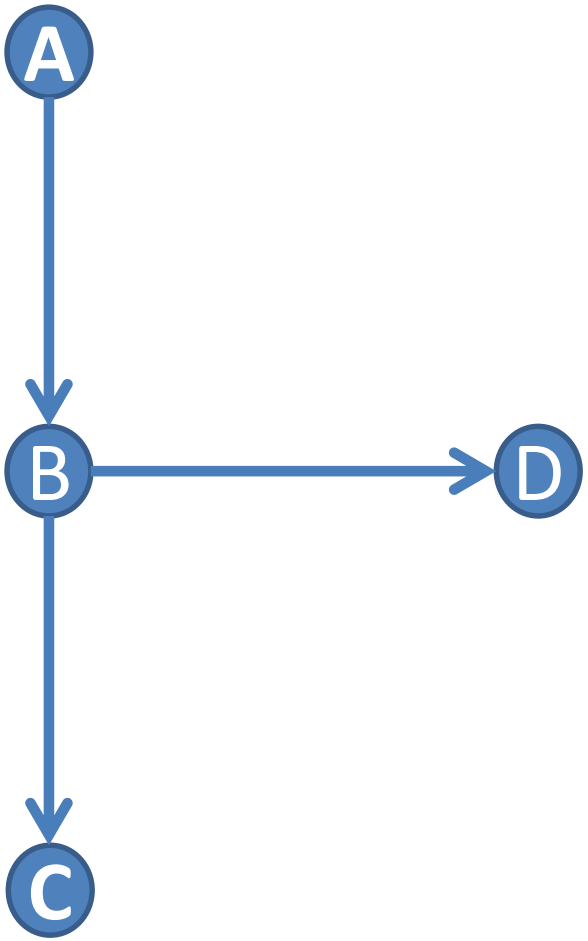


Bi-Parallel

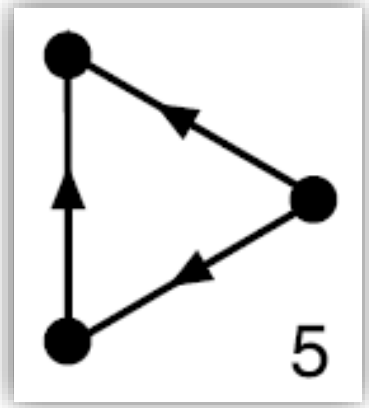
Exercise: Count the 3-node substructure

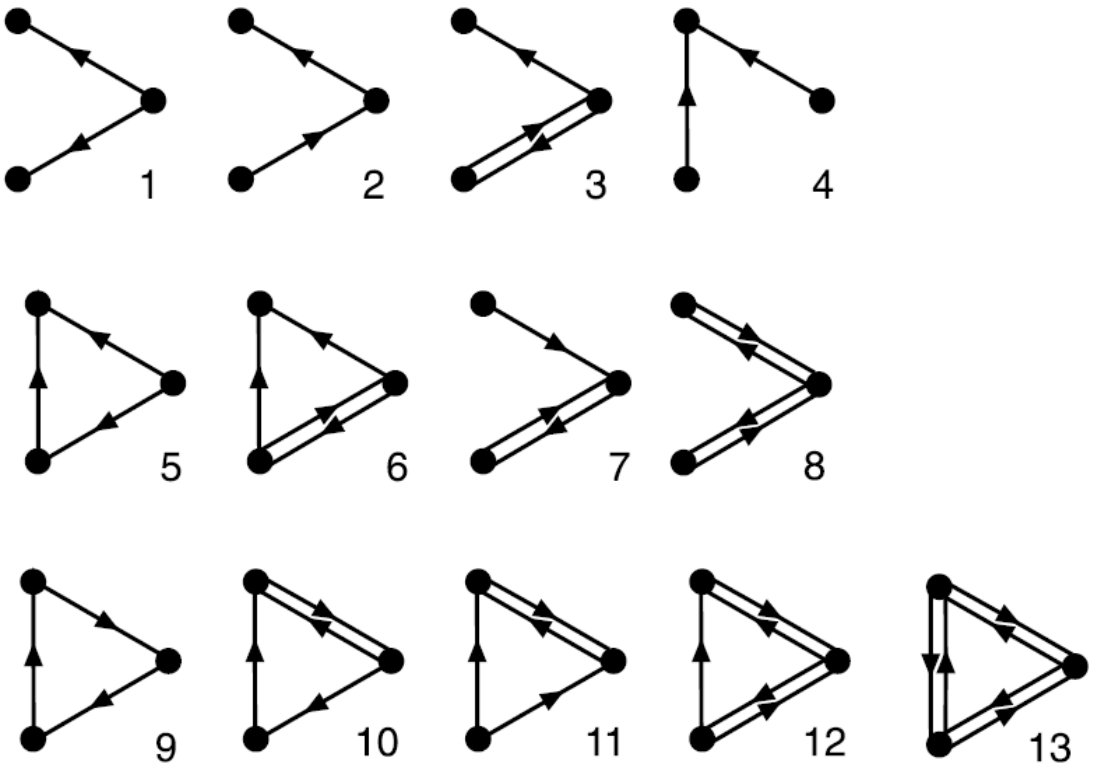


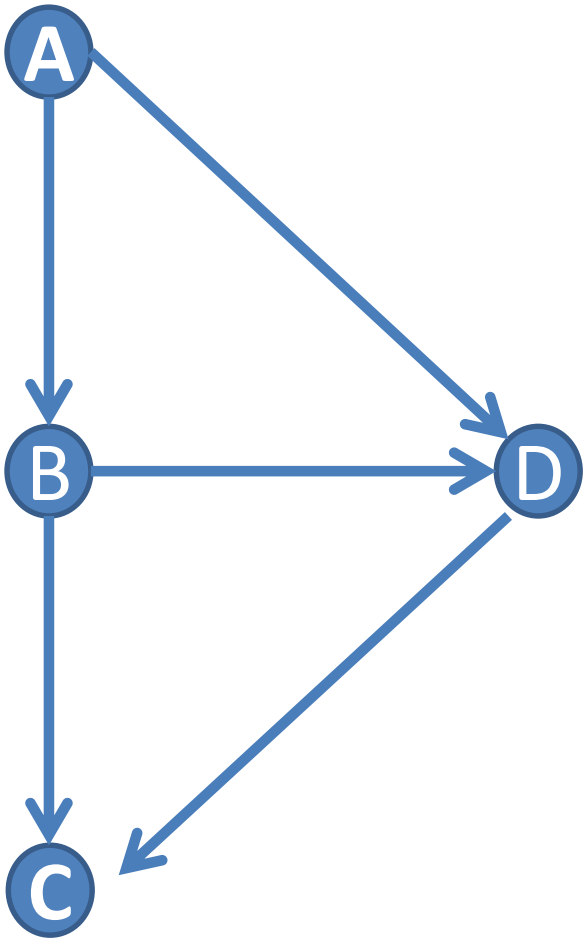




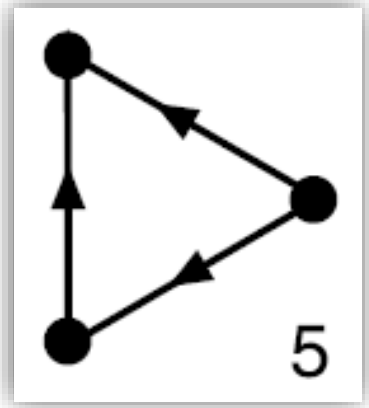
Exercise: Count the 3-node substructure

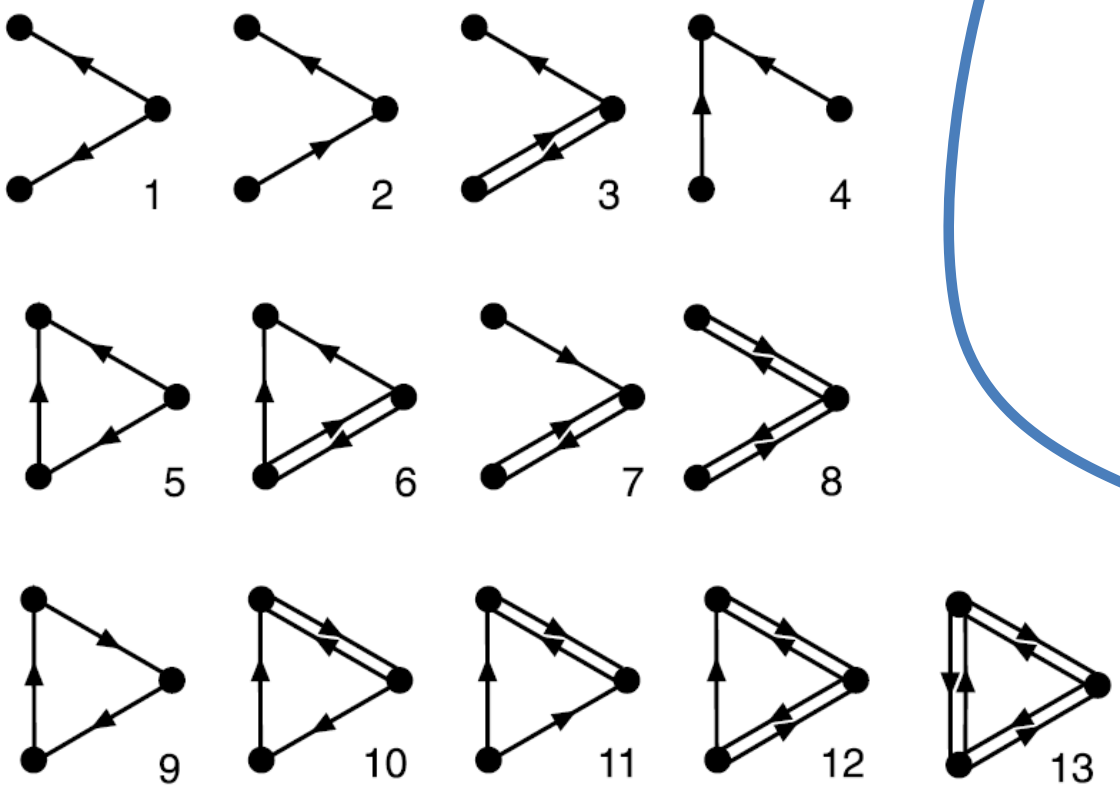


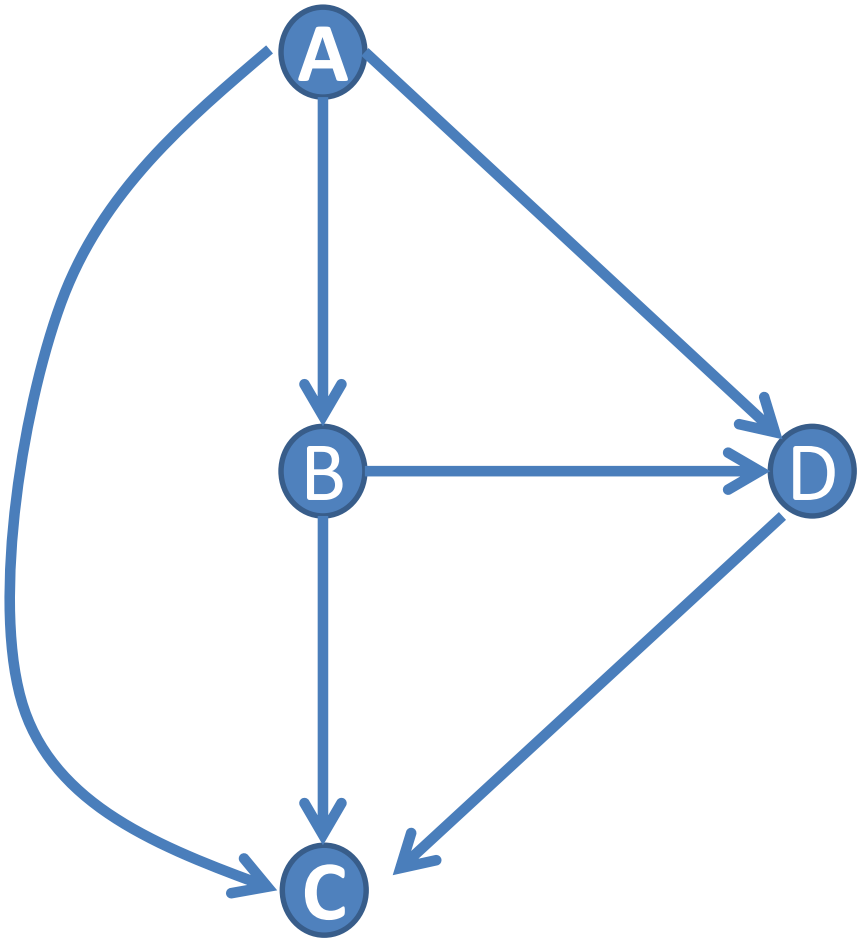
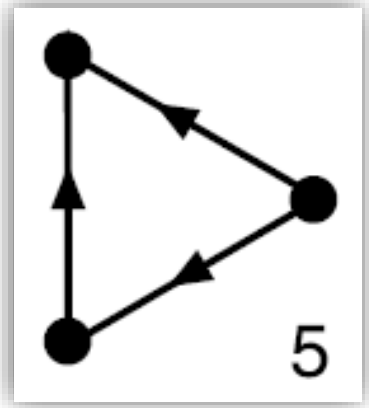
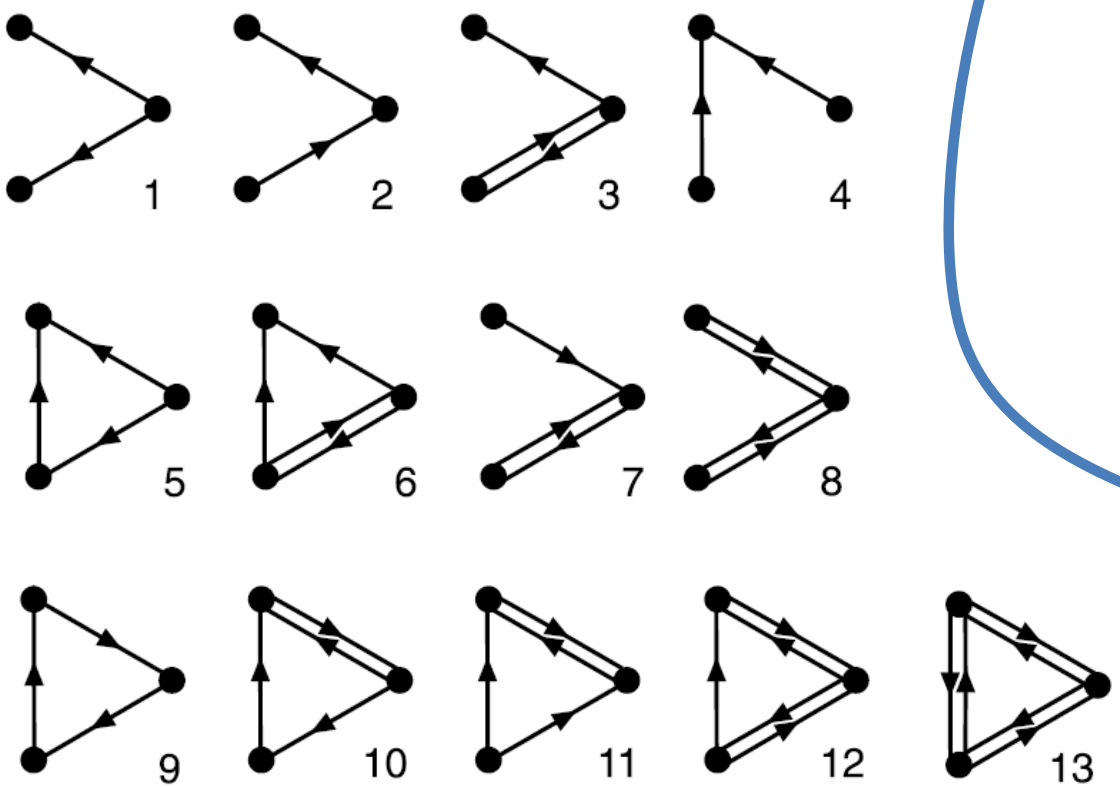
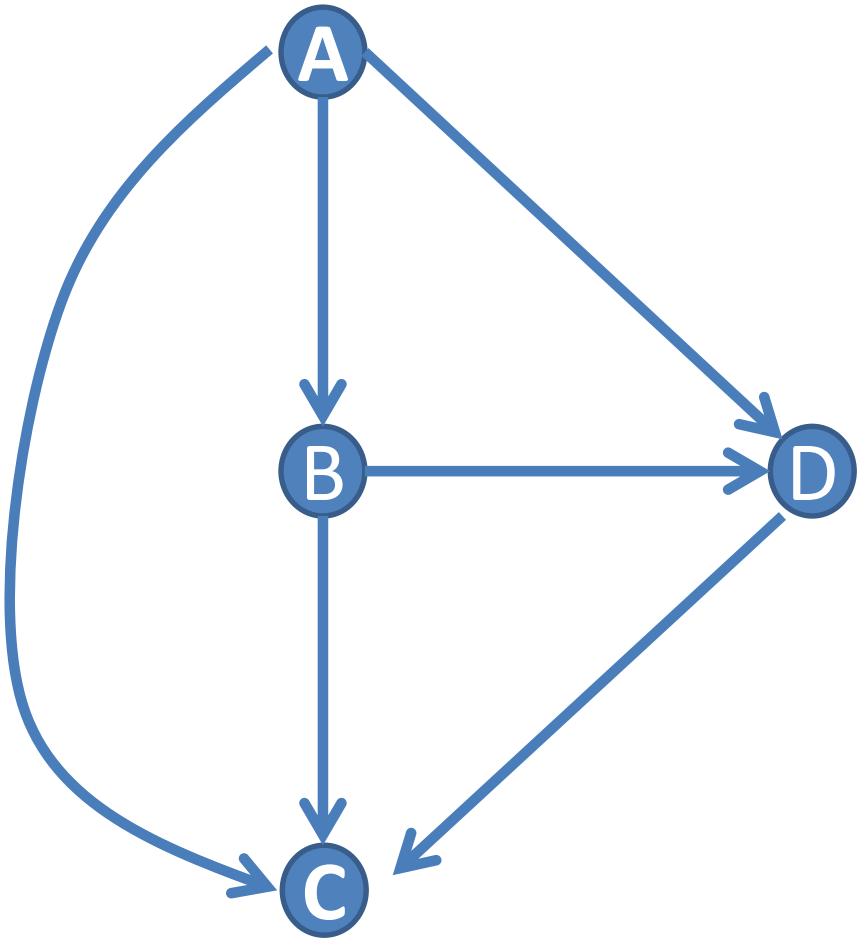




Exercise: Count the 3-node substructure







Exercise: Count the 3-node motifs